Hong Kong male subfertility links to mercury in human hair and fish.

The focus of the present study was on the relationship between Hong Kong male subfertility and fish consumption. Mercury concentrations found in the hair of 159 Hong Kong males aged 25-72 (mean age = 37 years) was positively correlated with age and was significantly higher in Hong Kong subjects than in European and Finnish subjects (1.2 and 2.1 ppm, respectively). Mercury in the hair of 117 subfertile Hong Kong males (4.5 ppm, $P < 0.05$) was significantly higher than mercury levels found in hair collected from 42 fertile Hong Kong males (3.9 ppm). Subfertile males had approx. 40% more mercury in their hair than fertile males of similar age. Although there were only 35 female subjects, they had significantly lower levels of hair mercury than males in similar age groups. Overall, males had mercury levels that were 60% higher than females. Hair samples collected from 16 vegetarians living in Hong Kong (vegans that had consumed no fish, shellfish or meat for at least the last 5 years) had very low levels of mercury. Their mean hair mercury concentration was only 0.38 ppm.